

Semester II

Course Code: CAM21T	Course Title: Data Structures
Course Credits: 03 (3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

On successful completion of this course, students will be able to:

1. Understand the basics of Data Structures.
2. Identify the appropriate data structures and algorithms for solving real world problems.
3. Understand the practical applications of Tree and Graph.
4. Understand the fundamentals of sorting and searching algorithms.

Course Contents:

Unit-1	
Introduction: Data Structure Definition, Basic Terminology and Concepts, Importance of Data Structures in Programming. Classification of Data Structures. Primitive Data Structures, Non-Primitive Data Structures. Stack: Definition, Memory Representation, Algorithms for Stack Operations (Push, Pop), Applications of Stack.	11 Hours
Unit-2	
Queue: Definition, Memory Representation, Linear Queue, Circular Queue, Enqueue, Dequeue. Applications of Queue. Linked Lists: Definition, Types. Singly Linked List: Implementation, Insertion [At the Beginning], Deletion [At the End]. Doubly Linked List: Memory Representation of Singly Linked List and Doubly Linked Lists. Applications of Linked List.	11 Hours
Unit-3	
Tree: Definition, Memory Representation Using Array and Linked List. Binary Tree: Definition, Traversal Algorithms [Pre-Order, In-Order, Post-Order], Construction of Tree from In-Order and Pre-Order, In-Order and Post-Order. Binary Search Trees: Insertion of a Node, Deletion of A Node. Advanced Tree Structures AVL And B-Trees: Definition and Applications.	11 Hours

Unit-4	
Graph: Definition, Memory Representation of Graph. Adjacency Matrix, Adjacency List. Graph Traversal Algorithms: Breadth-First Search (BFS), Depth-First Search (DFS). Sorting Techniques: Bubble Sort, Selection Sort [Algorithm, Time & Space Complexity]. Searching Techniques: Linear And Binary Search Sort [Algorithm, Time & Space Complexity]. Heap: Heap Operations and Applications.	11 Hours

Reference Books:

1. Data Structures Through C++ (4th Edition) Yashvant Kanetkar.
2. Data Structures and Algorithm Analysis in C++" by Mark Allen Weiss.
3. Data structure and Algorithms using C++ by Sachi Nandan Mohanty, Pabitra Kumar Tripathy.
4. Data Structures and Algorithms in C++, Second Edition by Adam Drozdek.

Course Code: CAM21P	Course Title: Data Structures using C++
Course Credits: 02 (0-0-2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Outcomes (COs):

On successful completion of this course, students will be able to:

1. Implement data structures using C++.
2. Demonstrate searching and sorting techniques using C++.
3. Demonstrate advanced programming skills through C++ programming language.

Laboratory Program List

Part A:

1. Program to find GCD of two numbers.
2. Program to implement Tower of Hanoi.
3. Program to print Fibonacci series.
4. Program to find largest and smallest element in an array.
5. Program to perform stack operations.
6. Program to perform Linear queue operations
7. Program to insert a node at the beginning of a singly linked list.
8. Program to delete a node at the end of a singly linked list.

Part B:

1. Program to construct a binary search tree
2. Program for Binary Tree traversal.
3. Program to implement DFS
4. Program to implement BFS
5. Program to Sort an Array (Selection Sort)
6. Program to Sort an Array (Bubble Sort)
7. Program to perform Linear Search of an Element in an Array.
8. Program to perform Binary Search of an Element in an Array.

Evaluation Scheme for Lab Examination [Marks: 40]

- **Writing:** One program from both Part A and Part B (15 Mark each) : $15 \times 2 = 30$
- **Execution:** Any one of the written Program: 05 Marks
- **Viva:** 05

Course Code: CAM22T	Course Title: Object Oriented Programming with Java
Course Credits: 03 (3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

On successful completion of this course, students will be able to:

1. Understand the Java programming fundamentals.
2. Describe with examples of basic Java OOP concepts.
3. Understand the Java Interfaces and Packages.
4. Deliberate the Details of Multithreading, Exception Handling & File Handling
5. Design GUI applications using tools like AWT.

Course Contents:

Unit-1	
Fundamentals of Object-oriented Programming: Object-oriented Paradigm, Basic Principles of Object-oriented Programming, Advantages of Object-Oriented Programming, Applications of Object-Oriented Programming. Introduction to Java Language: Java History, Features, Overview, Difference between C, C++ and Java, Java Environment- JDK, JVM, JRE and API, Java Program Structure, Java Tokens, Implementing a Java Program, Command Line Arguments. Java Programming Fundamentals: Data types, Variables & Constants, Keywords & Naming Conventions, Type Casting, Operators and Expressions, Control Structures, Jumping Statements.	11 Hours
Unit-2	
Classes & Objects Basics of Objects and Classes, Constructors, Access Modifiers, Method Overloading, Overloading Constructors, Static members, this keyword. Arrays : One dimensional Arrays, Two dimensional Arrays, Array of Objects. Strings: String Handling functions.	11 Hours

Unit-3	
Multithreading in Java Concepts of Thread, Thread Life Cycle, Creating Threads & Implementing Runnable Interface, Thread Synchronization & Thread Priority. Exception Handling Concepts of Exception, Different Types of Exceptions, Creating User-Defined Exceptions Using Try-Catch-Finally-Throw Blocks, Nested Try, Catch, Throw, and Throws Blocks.	11 Hours
Unit-4	
File Handling I/O Handling, I/O Streams, Types of Files, Byte Stream, Binary I/O Classes & Its Hierarchy, FileInputStream & FileOutputStream Classes, Object I/O Classes. Event Handling & GUI programming Event Handling, Event Types, Event Handling Mechanism, Keyboard & Mouse Handling, Introduction to AWT & GUI basics, AWT hierarchy of classes, AWT controls – Frames, Panels, Layout managers & other controls of AWT.	11 Hours

Reference Books:

1. D.S. Guru, M.T. Somashekara, & K.S. Manjunatha, Object Oriented Programming with Java, PHI Learning, 2017.
2. E Balagurusamy , Programming with JAVA, TMH, 2007
3. Herbert Schildt , Java 7, The Complete Reference, , 8th Edition, 2009

Course Code: CAM22P	Course Title: Programming with Java
Course Credits: 02 (0-0-2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Outcomes (COs):

On successful completion of this course, students will be able to:

1. Implement simple programs using Java Fundamental concepts.
2. Identify classes, objects, members of class and the relationships among them needed for finding the solution to specific problems using Objected Oriented Programming concepts of Java.
3. Design & Develop simple GUI programs using AWT GUI tool.

Laboratory Program List

Part A:

1. Program to find whether the given number is Positive, Negative or Zero.
2. Program to list the factorial of the numbers 1 to 10.
3. Program to demonstrate classes & objects.
4. Program to demonstrate method overloading.
5. Program to demonstrate single inheritance (simple calculator – base class, Advanced Calculator – derived class).
6. Program to find Maximum & Minimum element in one dimensional array of numbers.
7. Program to check whether the given string is palindrome or not.
8. Program to create a 'Student' class with Reg.no., name and marks of 3 subjects. Calculate the total marks of 3 subjects and create an array of 3 student objects & display the results.

Part B:

1. Program to generate negative array size exception
2. Program to generate NullPointerException.
3. Program that reads two integer numbers for the variables a and b. The program should catch NumberFormatException & display the error message.
4. Program to create AWT window with 4 buttons M/A/E/Close. Display M for Good Morning, A for Afternoon, E for evening and Close button to exit the window.
5. Program to demonstrate the various mouse handling events.
6. Program to read and write Binary I/O file.
7. Program to create window with three buttons father, mother and close. Display the

respective details of father and mother as name, age and designation using AWT controls.

8. Program to create menu bar and pull-down menus.

Evaluation Scheme for Lab Examination [Marks: 40]

- **Writing:** One program from both Part A and Part B (15 Mark each) : $15 \times 2 = 30$
- **Execution:** Any one of the written Program: 05 Marks
- **Viva:** 05

Course Code: CAM23T	Course Title: Operating Systems
Course Credits: 03 (3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

On successful completion of this course, students will be able to:

1. Understand the fundamentals of the operating system.
2. Describe the concepts of process, process management, CPU Scheduling, process synchronization, Dead locks, memory management and Virtual Memory management.
3. Illustrate the file system and structure.
4. Understand the UNIX OS, Shell Programming, Conditional Control Structures in Shell Programming.

Course Contents:

Unit-1	
Introduction: Definition, Computer System Components , User View , System View And System Goals , Batch Systems, Multi Programmed Systems, Time-Sharing Systems, Real-Time Systems, System Components, Operating System Services. Process: Process Concept, Process State Diagram Process Control Block, Process Scheduling- Scheduling Queues, Scheduler, Cooperating Process, Interprocess Communication. CPU Scheduling: Basic Concepts, Preemptive And Non-Preemptive Scheduling, Scheduling Criteria, Scheduling Algorithms-FCFS, Shortest Job First Priority Scheduling, Round Robin Scheduling.	11 Hours
Unit-2	
Process Synchronization: The Critical Section Problem, Solution For Critical Section Problem, Bakery Algorithm, Semaphores-Meaning, Types Of Semaphores, Synchronization Problems- Bounded Buffer Problem, Readers-Writers Problem . Deadlocks: Deadlock Characterization, Methods For Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery From Deadlock.	11 Hours

Unit-3	
Memory Management: Introduction, Logical Versus Physical Address Space, Dynamic Loading, Dynamic Linking, Swapping, Contiguous Allocation, Partitioned Memory Allocation, Paging, Virtual Memory Management-Segmentation, Segmentation With Paging. File System: File Concepts, File Attributes, File Operations, File Types, File Structure, Access Methods, Directorystructure, File-System Structure, Allocation Methods- Contiguous Allocation, Linked Allocation and Indexed Allocation, Free-Space Management.	11 Hours
Unit-4	
Introduction to Unix System: The Unix Operating System, The UNIX architecture. Shell Programming: Vi editor, shell types, shell command line processing, shell script features, executing a shell script, system and user-defined variables, expr command, shell screen interface, read and echo statement, command substitution, escape sequence characters, shell script arguments, positional parameters, test command, file test, string test, numeric test. Conditional Control Structures: if statement, case statement Looping Control Structure-while, until, for, statements. Jumping Control Structures – break, continue, exit. Shell Programs covering the above concepts.	11 Hours

Reference Books:

1. Operating System Concepts – 5th edition by Abraham Silberschartz and Peter Galvin, McGraw Hill,2000
2. Modern Operating Systems – Andrew S Tanenbaum, Prentice Hall
3. Operating Systems : Internals and Design Principles, William Stallings, Prentice Hall
4. Sumitabha Das: UNIX – Concepts and Applications, 4th Edition, Tata McGraw Hill, 2006.

Course Code: CAM23P	Course Title: Shell Programming
Course Credits: 02 (0-0-2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Outcomes (COs):

On successful completion of this course, students will be able to:

1. Develop skill in shell scripting to perform simple operations and problems.
2. Perform file manipulation using shell scripts.
3. Understand and implement shell scripts for system information.

Laboratory Program List

Part A:

1. Write a shell script to swap 2 values.
2. Write a shell script to check if the given number is even or odd.
3. Write a shell script to find the largest of 3 numbers.
4. Write a shell script to perform arithmetic operations.
5. Write a shell script to find the sum of first 10 natural numbers.
6. Write a shell script to display multiplication table of a given number.
7. Write a shell script to find the length of a given string.
8. Write a shell script to find factorial of a given number.
9. Write a shell script which counts the numbers of lines and number of words present in a given file.
10. Write a shell script to display the Fibonacci series upto N number.

Part B:

1. Write a shell script to search for particular element from an array of elements.
2. Write a shell script to calculate the TA, HRA and DA of an employee.
3. Write a shell script that displays a list of all files in the current directory to which the user has read write and execute permissions.
4. Develop an interactive script that asks for a word and file name and then tells how many times that word occurred in the file.
5. Write a shell script to extract a sub string from a given string.
6. Perform the following operations
7. Concatenate 2 strings
8. Rename a file
9. Delete a file

10. Copy the file
11. Write a shell script to display the
 - a) Version of the shell
 - b) The user information
 - c) Login date and time
 - d) List of processes running on the system
 - e) User home directory
12. Write a C program to display PID of parent and PID of child process.
13. Write a shell script that takes two filename as arguments. It should check whether the contents of two files are same or not, if they are same then second file should be deleted.
14. Assume a file with the given information
15. First Name Middle Name Age
16. -----
17. Write a shell script to
 - a. Sort the first name in alphabetical order
 - b. Sort the age in terms of ascending order
 - c. Sort the age in terms of descending order
 - d. Sort the middle name in alphabetical order

Evaluation Scheme for Lab Examination [Marks: 40]

- **Writing:** One program from both Part A and Part B (15 Mark each) : $15 \times 2 = 30$
- **Execution:** Any one of the written Program: 05 Marks
- **Viva:** 05

CIE, SEE and QP Pattern for Theory Courses

Total Lecture hours per paper: 44

No. of Units 4 (11 Hours Each)

Internal Assessment C1 = 10 Marks, C2 = 10 Marks

Semester End Theory Exam C3 = 80 Marks

Question paper pattern

Instructions: Answer Part-A and Part-B:

Part-A

Answer any 10 out of 12 Questions (3 Questions drawn from each unit). Each question carries 2 Marks. (10 X 2 =20)

Q. No. 1 to Q. No. 12.

Part-B

Answer all the Questions. Each question carries 15 Marks. (4 X 15 =60)

(Each question with internal choice and with maximum of 3 sub questions)